

PERIODIC TABLE & ELEMENT PROPERTIES

Conceptual Practice Workbook with Comprehensive Solutions

This workbook contains open-ended practice questions designed to test and reinforce your understanding of element classification and periodic trends. Each question is aligned with fundamental chemistry topics from the syllabus. Detailed step-by-step solutions are provided at the end of this document.

PART I: CONCEPTUAL PRACTICE PROBLEMS

Topic 1: Historical Milestones & Periodic Laws

Question 1 (Döbereiner's Triads): For a chemically similar triad of elements consisting of Lithium (Li, atomic mass = 7.0) and Potassium (K, atomic mass = 39.0), calculate the expected atomic mass of the middle element, Sodium (Na), using Döbereiner's Law of Triads. Explain the major limitation that led to the system's limited success.

Question 2 (Mendeleev & Newlands): State Newlands' Law of Octaves and Mendeleev's Periodic Law. Explain the significance of Mendeleev leaving blank spaces in his periodic table, specifically identifying the element discovered to fill the "eka-silicon" gap.

Question 3 (Moseley & Modern Law): State the mathematical formula resulting from Henry Moseley's work on X-ray spectra that established the physical basis of the Modern Periodic Law. Contrast the modern periodic law with Mendeleev's periodic law.

Topic 2: Structure of the Modern Periodic Table & Block Assignment

Question 4 (Block Assignment): An element has the ground-state electronic configuration $[\text{Ar}] 3d^6 4s^2$. Identify its block, group, and period in the periodic table, and state the rule used to assign its block.

Question 5 (f-block Constitution): Explain how the f-block is incorporated into the modern periodic table. Identify the two series of elements that make up the f-block and state where they are conventionally positioned.

Topic 3: Systematic IUPAC Nomenclature ($Z > 100$)

Question 6 (Systematic Names): Derive the systematic IUPAC names and three-letter symbols for elements with atomic numbers $Z = 115$ and $Z = 118$. List the specific numerical roots used in your derivation.

Topic 4: Atomic & Ionic Radii Principles

Question 7 (Isoelectronic Species): Define isoelectronic species. Arrange the ions Na^+ , Mg^{2+} , and Al^{3+} in order of decreasing ionic radius, and justify this order using the concept of effective nuclear charge.

Question 8 (Cation & Anion Expansion): Explain physically why a cation is always smaller than its parent atom ($r_{\text{cation}} < r_{\text{parent}}$) whereas an anion is always larger than its parent atom ($r_{\text{anion}} > r_{\text{parent}}$).

Question 9 (Period 2 Radii): Arrange the second-period elements **C**, **N**, **O**, and **F** in order of increasing atomic radius. Explain the fundamental physical mechanism driving this trend from left to right across a period.

Topic 5: Ionisation Enthalpy Trends & Orbital Exceptions

Question 10 (Subshell Stability Anomaly): Explain why the first ionisation enthalpy of Nitrogen is higher than that of Oxygen ($IE_1(\text{N}) > IE_1(\text{O})$), and why Beryllium's first ionisation enthalpy is higher than that of Boron ($IE_1(\text{Be}) > IE_1(\text{B})$). Support your answer with electronic configurations.

Question 11 (Successive Enthalpies): Define successive ionisation enthalpies and explain why the second ionisation enthalpy (IE_2) of any element is always higher than its first ionisation enthalpy (IE_1).

Question 12 (Period 3 Extremes): Among the elements Sodium (Na), Magnesium (Mg), Aluminium (Al), and Chlorine (Cl), identify the element with (a) the lowest first ionisation enthalpy, and (b) the highest first ionisation enthalpy. Explain your choices.

Topic 6: Electron Gain Enthalpy Trends & Anomalies

Question 13 (Chlorine vs Fluorine Anomaly): Chlorine has a more negative electron gain enthalpy than Fluorine ($\Delta_{\text{eg}} \text{H}(\text{Cl}) > \Delta_{\text{eg}} \text{H}(\text{F})$ in magnitude), even though Fluorine is smaller and more electronegative. Describe the electronic mechanism behind this anomaly.

Question 14 (Noble Gases & EGE Trend): Explain why noble gases exhibit large positive values of electron gain enthalpy ($\Delta_{\text{eg}} \text{H} > 0$). Additionally, arrange Nitrogen (N), Oxygen (O), and Fluorine (F) in order of increasing negative electron gain enthalpy.

Topic 7: Electronegativity & Chemical Character

Question 15 (Pauling Scale & Trends): Define electronegativity according to the Pauling scale. Explain how electronegativity is related to metallic character, and arrange C, N, O, and F in order of increasing electronegativity.

Topic 8: Metallic/Non-Metallic Character & Oxide Chemistry

Question 16 (Oxide Acidity Trend): Classify the following oxides as acidic, basic, or amphoteric: Na_2O , Al_2O_3 , SO_3 , K_2O , and BeO . Describe the trend in the acidity/basicity of oxides moving across Period 3 from left to right (Na to Cl).

Topic 9: Diagonal Relationship & First Element Anomalies

Question 17 (Charge/Size Ratio): Lithium shows similarities to Magnesium, and Beryllium shows similarities to Aluminium. Identify the name given to this phenomenon, and explain the two physical factors that cause diagonally adjacent elements to exhibit similar chemical properties.

Question 18 (First Element Anomalies): Explain why the first element of each main group (e.g., Lithium in Group 1, Beryllium in Group 2) shows chemical properties that deviate significantly from the rest of the members of their respective groups.

Topic 10: Miscellaneous Abundance & State Anomalies

Question 19 (Elemental Abundance): Identify the most abundant element in the Earth's crust by mass. Compare its abundance with Silicon, the second-most abundant element.

Question 20 (Mercury Liquid State): Why is Mercury (Hg) unusual among transition metals at room temperature? Provide the atomic-scale explanation (focusing on the 6s subshell and relativistic effects) for this unique physical property.

PART II: COMPLETE SOLUTIONS & EXPLANATIONS

Topic 1: Historical Milestones & Periodic Laws

Solution 1: According to Döbereiner's Law of Triads, the atomic mass of the middle element (Sodium, Na) should be the arithmetic mean of the atomic masses of the first (Lithium, Li) and third (Potassium, K) elements:

$$\text{Atomic Mass (Na)} \approx [\text{Atomic Mass (Li)} + \text{Atomic Mass (K)}] / 2 = [7.0 + 39.0] / 2 = \mathbf{23.0 \text{ u}}$$

Major Limitation: This classification system had limited success because it was **applicable only to a few sets of elements** known at the time. Most elements could not be grouped into neat triads, and the relationship failed for heavier elements.

Solution 2: Newlands' Law of Octaves states that when elements are arranged in order of increasing atomic mass, **every eighth element** exhibits physical and chemical properties analogous to the **first element** (similar to octaves in music).

Mendeleev's Periodic Law states that the properties of elements are a periodic function of their **atomic masses**.

Mendeleev left empty gaps in his table because he recognized that some elements were undiscovered. By analyzing periodic trends, he predicted their properties. His predicted element "eka-silicon" was later discovered and identified as **Germanium (Ge)**, which matched his predictions almost perfectly, validating his periodic system.

Solution 3: Henry Moseley studied the frequency of X-rays emitted by bombardments of metals and plotted the square root of the frequency ($\sqrt{\nu}$) against the atomic number (Z), producing a straight line represented by:

$$\sqrt{\nu} = a(Z - b) \text{ (where } a \text{ and } b \text{ are constants).}$$

This work led to the Modern Periodic Law: the physical and chemical properties of elements are a periodic function of their **atomic numbers (Z)**. This resolved anomalies of Mendeleev's law (which was based on **atomic mass**) because atomic number reflects the true proton count and nuclear charge, which directly dictate electronic configuration and chemical behavior.

Topic 2: Structure of the Modern Periodic Table & Block Assignment

Solution 4: For the configuration $[\text{Ar}] 3d^6 4s^2$:

1. **Block:** Since the differentiating (last) electron enters the 3d orbital, the element belongs to the **d-block**.
2. **Period:** The highest principal quantum number is $n = 4$ (from the 4s orbital), so it belongs to **Period 4**.
3. **Group:** For d-block elements, $\text{Group} = (\text{number of d-electrons} + \text{number of s-electrons}) = 6 + 2 = \mathbf{\text{Group 8}}$ (Iron, Fe).

Rule: An element is assigned to a block based on the subshell that receives the final valence electron during Aufbau filling.

Solution 5: The f-block consists of elements whose valence electrons fill the f-orbitals (specifically the $(n-2)f$ subshell). To maintain a clean 18-column width for the main periodic table, the f-block is placed in two separate rows at the very bottom:

1. **Lanthanides (4f series):** 14 elements from Cerium ($Z=58$) to Lutetium ($Z=71$), belonging to Period 6.
2. **Actinides (5f series):** 14 elements from Thorium ($Z=90$) to Lawrencium ($Z=103$), belonging to Period 7.

All f-block elements formally belong to **Group 3** of the periodic table.

Topic 3: Systematic IUPAC Nomenclature ($Z > 100$)

Solution 6: The systematic names are derived from numerical roots: 0 = *nil*, 1 = *un*, 2 = *bi*, 3 = *tri*, 4 = *quad*, 5 = *pent*, 6 = *hex*, 7 = *sept*, 8 = *oct*, 9 = *enn*, followed by the suffix *-ium*.

1. **Z = 115 (Moscovium):** Root for 1 is *un*, 1 is *un*, 5 is *pent*. Systematic name: *un + un + pent + ium* = **Ununpentium**. Symbol: **Uup** (official symbol: Mc).

2. **Z = 118 (Oganesson):** Root for 1 is *un*, 1 is *un*, 8 is *oct*. Systematic name: *un + un + oct + ium* = **Ununoctium** (note: elide one 'i' in oct-ium). Symbol: **Uuo** (official symbol: Og).

Topic 4: Atomic & Ionic Radii Principles

Solution 7: Isoelectronic species are atoms and ions that contain the **same number of electrons**. Na^+ , Mg^{2+} , and Al^{3+} are all isoelectronic, each possessing 10 electrons ($1s^2 2s^2 2p^6$ configuration).

The size of isoelectronic species is inversely proportional to their atomic number / nuclear charge (Z):

Ionic Radius $\propto 1/Z$

The nuclear charges (Z) are Na ($Z=11$), Mg ($Z=12$), and Al ($Z=13$). Because Al^{3+} has the highest nuclear charge (13 protons pulling on 10 electrons), it has the strongest pull on the electron cloud, contracting it the most. Na^+ has the weakest nuclear pull (11 protons).

The decreasing order of size is: $\text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+}$.

Solution 8:

1. **Cations** are smaller than their parent atoms because the removal of one or more valence electrons often results in the loss of an entire outer electronic shell, and increases the **effective nuclear charge per remaining electron**. The remaining electrons experience a stronger net electrostatic attraction toward the nucleus, contracting the radius.

2. **Anions** are larger than their parent atoms because adding electrons to the same energy shell increases **inter-electronic repulsion** among the valence electrons. To minimize this electrostatic repulsion, the electron cloud expands outward while the nuclear charge (proton count) remains completely constant.

Solution 9: The increasing order of atomic radius for the second-period elements is: **F < O < N < C**.

Mechanism: Moving from left to right across a period, the atomic number (nuclear charge, Z) increases, but the extra valence electrons are added to the **same principal energy shell** ($n = 2$). These electrons shield each other very poorly from the nucleus. Consequently, the **effective nuclear charge (Z_{eff})** increases steadily across the period, pulling the outer electron cloud closer to the nucleus and causing a steady decrease in atomic size.

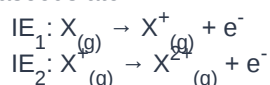
Topic 5: Ionisation Enthalpy Trends & Orbital Exceptions

Solution 10:

1. **Nitrogen vs Oxygen:** Nitrogen has the configuration $1s^2 2s^2 2p^3$, while Oxygen is $1s^2 2s^2 2p^4$. Nitrogen's 2p subshell is exactly **half-filled**, which possesses exceptional thermodynamic stability due to symmetrical distribution and maximum exchange energy. Oxygen has a paired electron in one of its 2p orbitals; the resulting inter-electronic repulsion makes it much easier to remove the first electron from Oxygen.

2. **Beryllium vs Boron:** Beryllium is $1s^2 2s^2$ (fully filled, stable 2s subshell), while Boron is $1s^2 2s^2 2p^1$. Removing an electron from Beryllium requires overcoming a stable subshell and extracting a 2s electron, which is more penetrating and closer to the nucleus than Boron's lone 2p electron. Boron's 2p electron is highly shielded by the inner 2s electrons, making it easier to remove.

Solution 11: Successive ionisation enthalpies refer to the energy required to remove electrons sequentially from a gaseous atom:



IE_2 is always greater than IE_1 because the second electron is removed from a positively charged cation (X^+). The

attractive electrostatic force of the nucleus holds the remaining electrons much more tightly in a cation, requiring significantly more energy to overcome.

Solution 12:

(a) **Lowest IE_1 is Sodium (Na):** Sodium belongs to Group 1 (alkali metals) and has a valence configuration of $3s^1$. It has the largest atomic size in Period 3 and its lone valence electron is highly shielded from the nucleus, allowing it to be removed very easily to form a highly stable, noble-gas-like octet core ([Ne]).

(b) **Highest IE_1 is Chlorine (Cl) [excluding Argon]:** Chlorine has the smallest atomic radius and highest effective nuclear charge (Z_{eff}) among the active elements of Period 3. The outer valence shell ($3s^2 3p^5$) is extremely close to the nucleus, resulting in an exceptionally strong electrostatic attraction that requires immense energy to break.

Topic 6: Electron Gain Enthalpy Trends & Anomalies

Solution 13: Fluorine's valence configuration is $2s^2 2p^5$, while Chlorine's is $3s^2 3p^5$. Fluorine is an extremely small atom, and its 9 electrons are packed into a very compact, small **2p subshell**. When an extra electron is added to Fluorine, it experiences massive **inter-electronic repulsion** from the existing dense electron cloud, which partially offsets the nuclear attraction. Chlorine's valence shell is the larger, roomier **3p subshell**, where the added electron experiences minimal inter-electronic repulsion. Consequently, Chlorine has a more negative electron gain enthalpy than Fluorine (Cl = -349 kJ/mol vs F = -328 kJ/mol). Chlorine has the most negative electron gain enthalpy of all elements.

Solution 14: Noble gases have completely filled valence subshells ($ns^2 np^6$), which represents an extremely stable closed-shell configuration. There are no vacant low-energy orbitals available to accommodate an extra electron. Any incoming electron must be placed into a completely new, higher energy principal quantum shell ($n+1$), which is highly unstable and requires putting external energy into the system (endothermic process, $\Delta_{\text{eg}} H > 0$).

The increasing order of negative electron gain enthalpy is: **N < O < F** (Nitrogen has a half-filled shell and has a near-zero/positive enthalpy; Fluorine has the most negative value of the three).

Topic 7: Electronegativity & Chemical Character

Solution 15: Electronegativity (χ) is a dimensionless chemical property representing the relative ability of a bonded atom in a molecule to attract the shared pair of electrons in a covalent bond toward itself. On the Pauling scale, Fluorine is defined as the most electronegative element with a value of 4.0.

Relation to Metallic Character: Electronegativity is inversely related to metallic (electropositive) character. As electronegativity increases, the tendency to lose electrons easily (forming cations) decreases, which means the element's metallic character **decreases**.

The increasing order of electronegativity for the specified Period 2 elements is: **C < N < O < F**.

Topic 8: Metallic/Non-Metallic Character & Oxide Chemistry

Solution 16:

1. Classifications:

- Basic Oxides: **Na_2O** and **K_2O** (highly electropositive metals, react with water to form strong bases).
- Acidic Oxides: **SO_3** (highly electronegative non-metal, reacts with water to form strong sulphuric acid).
- Amphoteric Oxides: **Al_2O_3** and **BeO** (react with both acids and bases to form salt and water).

2. **Period 3 Trend (L → R):** As we move from left to right across Period 3, the electropositivity of the elements decreases and their electronegativity (non-metallic character) increases. Consequently, the nature of their oxides transitions systematically from strongly basic to strongly acidic:

Na_2O (Strongly Basic) → MgO (Basic) → Al_2O_3 (Amphoteric) → SiO_2 (Weakly Acidic) → P_4O_{10} (Acidic) → SO_3 (Acidic) → Cl_2O_7 (Strongly Acidic).

Topic 9: Diagonal Relationship & First Element Anomalies

Solution 17: The phenomenon is called the **diagonal relationship**. It is observed between certain elements of Period 2 (Groups 1, 2, 13) and their diagonally adjacent neighbors in Period 3 (Groups 2, 13, 14).

The two primary physical criteria that cause this similarity are:

1. **Comparable atomic and ionic sizes** (e.g., $r(\text{Li}^+) \approx 76 \text{ pm}$, $r(\text{Mg}^{2+}) \approx 72 \text{ pm}$).
2. **Similar charge densities / ionic potential (ϕ)**, defined as the ratio of ionic charge to ionic radius (Charge / Size). Because they have similar polarizing power, they form compounds with highly similar covalent characters and solubilities.

Solution 18: The anomalous behavior of the first (lightest) element of each main group is driven by three key factors:

1. **Unusually small atomic and ionic size**, which results in an exceptionally high surface charge-to-size ratio (charge density).
2. **Unusually high electronegativity and ionisation enthalpy** compared to heavier group members.
3. **Non-availability of d-orbitals in their valence shell ($n = 2$)**. Unlike heavier members which can expand their octet using vacant d-orbitals (coordination number up to 6), Period 2 elements are strictly capped at a maximum coordination number of 4 using only their 2s and 2p orbitals (four valence orbitals).

Topic 10: Miscellaneous Abundance & State Anomalies

Solution 19: Oxygen (O) is the most abundant element in the Earth's crust by mass, accounting for approximately **46.6%** of its composition.

It is followed by **Silicon (Si)** as the second-most abundant element at approximately **27.7%**. Together, silicates and aluminosilicates formed from Oxygen and Silicon constitute the vast majority of minerals in the continental and oceanic crusts.

Solution 20: Mercury (Hg, $Z=80$) is unique among transition metals as it exists as a **liquid at room temperature** (melting point $\approx -39^\circ\text{C}$) and is highly volatile.

Atomic-Scale Explanation: Mercury has the closed-shell configuration $[\text{Xe}] 4f^{14} 5d^{10} 6s^2$. Because of its extremely high nuclear charge ($Z=80$), the inner electrons travel at speeds approaching a significant fraction of the speed of light. This induces a relativistic effect that causes a massive increase in mass and a subsequent **relativistic contraction of the outermost 6s orbital**.

This relativistic contraction pulls the $6s^2$ electron pair extremely close to the nucleus, shielding them tightly. As a result, the $6s^2$ electrons behave almost like noble gas electrons and are highly inert. They do not participate effectively in metallic bonding with adjacent mercury atoms. The weak metallic bonding (van der Waals-like cohesive forces) is easily overcome by thermal energy at room temperature, causing Mercury to remain in the liquid state.